VIRGINIA STATE UNIVERSITY · TRI-CITIES REGION

A Semester-Long Toe-Dip Into IoT and AI-Assisted Building

For business students with no engineering background. Not a certification track — a fast, guided first contact with how physical devices, data, and AI-assisted software actually fit together in today's economy.

Working program name · Spring 2027 cohort · Draft overview — schedule subject to revision

DURATION	FORMAT	AUDIENCE	OUTCOME
14 weeks + finals	**2 sessions/week, hands-on hardware labs**	**Business students · no CS/EE prerequisite**	**Prototype, pitch deck, portfolio, Career Day**

How Students Learn to Direct AI, Not Just Use It

Every real task in the course splits the way real AI-assisted work does — a judgment call, then a mechanical build. Both halves are taught with a named framework, not left implicit.

JUDGMENT STEP

4D Framework

`Delegation` `Description` `Discernment` `Diligence`

BUILD STEP

RTFC Framework

`Role` `Task` `Format` `Context`

Turning a settled decision into working firmware or code, by

Deciding what to build, and whether a design or a wiring plan is actually safe to ship.

directing AI with a clear prompt.

The 14-Week Arc

JAN → MAY 2027

Four phases, each ending in something real: a validated problem, a working device, a live product, a delivered pitch.

PHASE 1 FOUNDATIONS Weeks 1–4

W1 **Course Kickoff**

Set up AI tools, build a public portfolio, publish a first post.

Portfolio Live

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W2 **Understanding Systems & IoT**

Explore real use cases, pick a focus, tinker with a device.

Use-Case Report

VOLTPOP

W3 **Problem & User Discovery**

Validate the problem, build a persona and journey map.

Persona & Journey Map

BUILDER TECH

W4 **Business Models & Market Fit**

Generate and evaluate ideas, lock a direction, launch a brand.

Business Model & Brand

EXPLAY

PHASE 2 BUILD THE PROTOTYPE Weeks 5–8

W5 **Hardware Setup & Sensor Basics**

Wire an ESP32 and sensor, write firmware by AI prompt.

Working Hardware + Firmware

VOLTPOP

W6 **Designing the User Experience**

Map the future user flow, prototype the key screens.

Low-Fidelity Prototype
BUILDER TECH

W7 **Data Flow & Systems Architecture**

Map sensor-to-app data flow; prove hardware and software actually talk.

End-to-End Integration Test
VOLTPOP

SPRING BREAK — NO CLASSES

W8 **Full Build — Iterations 1 & 2**

Build the real app; integrate hardware and software end-to-end.

Iterations 1 & 2
VOLTPOP

PHASE 3 ITERATE & LAUNCH Weeks 9–10

W9 **Full Build — Iterations 3 & 4**

Apply feedback, bring in security concepts, finish the build.

Final Iteration
EXPLAY

W10 **Testing & Going Live**

Publish, plan go-to-market, launch publicly with a demo video.

Live Product + Demo Video
EXPLAY

PHASE 4 PITCH & PRESENT Weeks 11–14

W11 **Crafting Your Pitch**

Structure the story, build the deck.

Pitch Deck Draft
EXPLAY

W12 **Practice & Feedback**

Rehearse live, get judged feedback, revise.

Rehearsal + Video
EXPLAY

BUFFER WEEK — POLISH, NO NEW CONTENT

W14

Final Presentations & Career Day

Live presentations, finished portfolio, resume — in front of regional employers.

Final Pitch + Career Day

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Built On Tools Students Can Keep Using

Nearly every tool in the course is free or open-source. The budget funds people and hands-on time — not software licenses.

- Claude Code
- GitHub
- ESP32 + Sensors
- Lovable
- Canva

Who's Teaching It

Javon Guerrier

BUILDER TECH

Program Director; Business Intelligence & Design instruction.

Andrew "Drew" Foulks

VOLTPOP

Computer Science SME; Engineering & Computer Science instruction.

Emanuel Perez

EXPLAY

Entrepreneurship & Innovation instruction and program design.

Dr. Shawn Nicholson

INSTITUTIONAL LIAISON

Operations Director; liaison to Virginia State University.

Prepared for stakeholder discussion — Virginia State University, Tri-Cities region. Program name shown is a working title, not yet finalized.

Curriculum current as of August 2026. Schedule and deliverables are subject to revision before launch.